

Weather API Service – API Documentation

Overview

The **Weather API Service** is a simple REST API built with **FastAPI** that integrates with the [OpenWeatherMap API](#).

It provides endpoints to:

- Fetch **current weather** by **city name** or **coordinates**
- Fetch **forecast** (1–5 days) by **city name** or **coordinates**
- Maintain **search history** (stored in memory)
- Apply **rate limiting** rules to prevent abuse

Additionally, it includes a **minimal HTML frontend** for testing and demonstration.

Authentication

- Requires an **OpenWeatherMap API key**.
- You must sign up at <https://openweathermap.org> and generate an API key.
- Replace `API_KEY` in the code with your key, or load it from an `.env` file.

Base URL

When running locally:

`http://127.0.0.1:8000`

Endpoints

1. Current Weather

Fetches current weather by **city name** or **geographic coordinates**.

Endpoint:

`GET /weather`

Query Parameters:

Parameter	Type	Required	Description
city	string	optional	City name (e.g., Nairobi)
lat	float	optional	Latitude (e.g., -1.29)
lon	float	optional	Longitude (e.g., 36.82)

Either city OR both lat(latitude) and lon(longitude) are required.

Example Requests:

`GET /weather?city=Nairobi`

`GET /weather?lat=-1.29&lon=36.82`

Response (200):

```
{
  "coord": { "lon": 36.82, "lat": -1.29 },
  "weather": [{ "description": "scattered clouds" }],
  "main": { "temp": 24.5, "humidity": 60 },
  "name": "Nairobi"
}
```

Errors:

- 400 Bad Request – if neither city nor coordinates are provided.
- 429 Too Many Requests – if requesting the same city again within 5 minutes.
- 401 Unauthorized – if API key is invalid.

2. Forecast

Fetches weather forecast for **1–5 days** by **city name** or **coordinates**.

Uses OpenWeatherMap's 3-hour interval forecast (cnt = days*8).

Endpoint:

GET /weather/forecast

Query Parameters:

Parameter	Type	Required	Description
city	string	optional	City name (e.g., Nairobi)
lat	float	optional	Latitude (e.g., -1.29)
lon	float	optional	Longitude (e.g., 36.82)
days	int	yes	Number of days (1–5)

Example Requests:

GET /weather/forecast?city=Nairobi&days=3

GET /weather/forecast?lat=-1.29&lon=36.82&days=2

Response (200):

```
{
  "city": { "name": "Nairobi" },
  "list": [
    {
```

```
"dt": 1694376000,  
"main": { "temp": 23.5 },  
"weather": [{ "description": "light rain" }]  
},  
...  
]  
}
```

Errors:

- 400 Bad Request – if invalid input or days not between 1–5.
- 429 Too Many Requests – if requesting the same city again within 5 minutes.
- 401 Unauthorized – if API key is invalid.

3. Search History

Retrieve all past weather and forecast searches.

Endpoint:

GET /weather/history

Response (200):

```
[  
  { "query": "Nairobi", "timestamp": "Wed Sep 10 21:05:43 2025", "type": "current" },  
  { "query": "-1.29,36.82", "timestamp": "Wed Sep 10 21:06:10 2025", "type": "forecast" }  
]
```

4. Clear History

Clear all stored search history.

Endpoint:

DELETE /weather/history

Response (200):

```
{ "detail": "History cleared" }
```

5. Frontend UI

A simple browser-based UI to test the API.

Endpoint:

GET /

Features:

- City input form for current weather
- Button for 3-day forecast
- View search history
- Clear all history button

Rate Limiting Rules

- **Current Weather (/weather)**
 - Each IP can only request the **same city/coordinates once every 5 minutes**.
 - Different cities/coordinates are allowed immediately.
- **Forecast (/weather/forecast)**
 - First forecast request for a city is allowed immediately after a weather search.
 - Further forecast requests for the same city are blocked for 5 minutes.
 - Different cities are allowed immediately.

If violated, the API returns:

```
{
  "detail": "Too many weather requests for Nairobi. Please wait 5 minutes."
}
```

Data Flow

1. Client calls /weather or /weather/forecast with city/coordinates.
2. API checks **rate limiting rules**.
3. If allowed → forwards request to OpenWeatherMap.
4. Response saved in **in-memory search history**.
5. Result returned to client.

Limitations

- Search history is stored in memory → cleared when app restarts.
- API key is hardcoded unless moved to .env.
- Free OpenWeatherMap tier has **60 requests/minute** limit.
- No database persistence.

Example Workflow

1. User searches current weather in Nairobi:
GET /weather?city=Nairobi
→ Returns weather, logs to history.

2. Immediately requests forecast for Nairobi:
GET /weather/forecast?city=Nairobi&days=3
→ Allowed (first forecast request).
3. Repeats forecast for Nairobi within 5 minutes:
→ Blocked (429 Too Many Requests).
4. Tries forecast for Mombasa:
→ Allowed (different city).s